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Comparison of near-infrared and Raman spectroscopy for the determination of the density of polyethylene pellets

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ABSTRACT

In this study, we compare near-infrared (NIR) and Raman spectroscopy for the determination of the density of linear low density polyethylene (PE) (in a pellet form). As generally known, Raman spectral features are more selective than those of NIR for most chemical samples. NIR spectroscopy has been more extensively used for the quantitative analysis of polymers, but Raman spectroscopy is the better choice as long as the problem of reproducibility of Raman measurements (especially for solid samples), mostly resulting from insufficient sample representation due to probing only localized chemical information and the sensitivity of sample placement with regard to the focal plane, can be overcome. To improve sample representation and reproducibility of Raman measurements, we have employed the wide area illumination (WAI) Raman scheme, capable of illuminating a laser onto a large sample area (28.3 mm^2) for Raman spectral collection (a 6-mm laser spot with a focal length of 248 mm). Diffuse reflectance NIR spectra of PE pellets were collected using a sample moving system which allowed for the scanning of large areas. The prediction error was 0.0008 g cm^{-3} for Raman spectroscopy and 0.0011 g cm^{-3} for NIR spectroscopy. The harmonization of inherently selective Raman features and a reproducible spectral collection with correct sample representations using the WAI scheme led to an accurate determination of the density of the PE pellets.
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1. Introduction

Near-infrared (NIR) spectroscopy has been extensively studied for quantitative analysis of various properties of diverse polymers since it is simple, fast and non-destructive [1–7]. When polymer samples are solid (such as pellets or flakes), diffuse reflectance is practically the only choice for obtaining NIR spectra while large samples are rotating or reciprocating to ensure spectral reproducibility. Transmission type spectral collection was also employed for the monitoring of molten polymers in an extruder [5,6]. Although the NIR spectral features are not highly distinct, the practical advantages of NIR for routine quantitative analysis make it one of the most adopted spectroscopic methods in polymer analysis, especially for quantitative analysis combined with chemometrics.

In the field of polymer research, Raman spectroscopy has been very effective and widely used to elucidate structural changes of polymers [8]. Additionally it is also capable of collecting spectra quickly and provides more selective spectral features than NIR spectroscopy; however, Raman spectroscopy has been less commonly utilized for quantitative analysis of polymers. Several researches have been conducted on the use of Raman spectroscopy to determine properties of polymers such as density and crystallinity [7,9–11]. Usually polymer samples are in large granule forms, such as pellets or flakes, which means that the reproducibility of Raman measurements could be degraded by the fact that the laser illuminates only a small spot when collecting spectral information, thereby increasing the potential for inaccurate compositional sample representation. In addition, Raman intensity could be sensitive to the variations of surface morphology and sample orientation. Because of this, Furukawa et al. reported that the accuracy for the determination of ethylene content in polypropylene using Raman spectroscopy was inferior to that of using NIR spectroscopy [7]. They explained that probing a small volume was the major cause of the degraded quantitative performance using Raman spectroscopy.

To fully take advantage of the superior spectral selectivity of Raman spectroscopy, it is necessary to devise a proper spectral collection scheme capable of representing correct sample identities with good reproducibility. The use of a wide area illumination (WAI) Raman scheme, capable of illuminating a laser onto a large sample area (28.3 mm^2) for Raman spectral collection (a 6-mm laser spot with a focal length of 248 mm), has improved the sample representation and reproducibility of Raman measurements in the analysis of various pharmaceutical products [12–15] and in-line monitoring of wet granulation processes [16].

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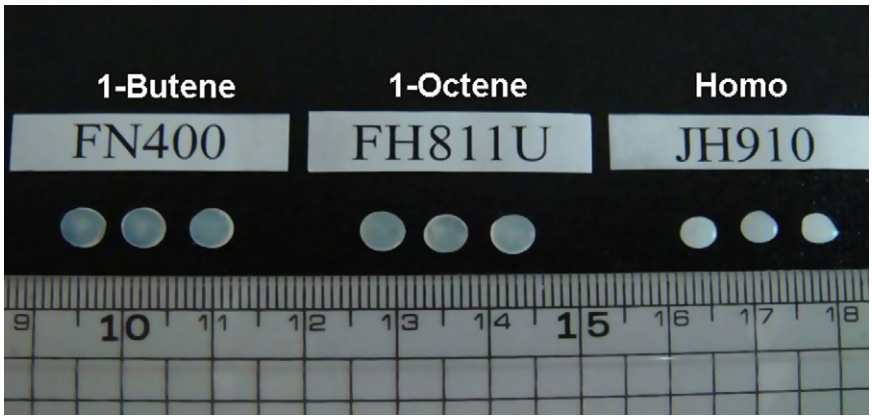

Fig. 1. Picture of the three different grades of pellets (centimeter scale).

In this research, we compared the quantitative performance of NIR and Raman spectroscopy in measuring the density of polyethylene (PE) pellets (picture shown in Fig. 1). The density of polymer is governed by its molecular arrangement during polymerization. The more ordered molecular arrangement (the more compact molecular packing in a given volume) leads to increase of density as well as crystallinity; while, the lower density is resulted from more disordered molecular arrangement. NIR spectra were collected in a diffuse reflectance configuration by reciprocating large amounts of PE pellets during spectral collection. The WAI scheme was used to collect Raman spectra without sample rotation or reciprocation. Since the size of the pellets was relatively large, as shown in Fig. 1, they required a Raman collection scheme that covered a wide area. Using partial least squares (PLS) [17], calibration models were developed using both NIR and Raman spectral datasets, and the corresponding results were compared each other.

2. Experimental

2.1. Sample preparation

Twenty-five polyethylene (LLDPE: linear low density polyethylene) pellets were obtained from a petrochemical company in Korea. The whole samples were composed of three different grades: 1-butene (12 samples), 1-octene (9 samples) and home (4 samples). The density of the samples ranged from 0.918 to 0.961 g cm^{-3}. Fig. 1 shows pictures of the three different grades of pellets (the scale is centimeter). The approximate ranges of the width and thickness of samples were 3.0–4.5 and 2.0–3.0 mm, respectively. The shapes of the 1-butene and 1-octene grade pellets were disk-like, while the homo grade pellets were relatively more spherical. The 1-butene pellets were relatively translucent while the home grade pellets were relatively opaque, with the 1-octene pellets in the middle.

2.2. NIR spectral collection

NIR spectra were collected using a Foss NIRSystems model 6500 spectrometer equipped with a quartz halogen lamp and four PbS detectors to collect diffused radiation from the samples. The spectrometer was equipped with a sample moving system which allowed scanning of large areas of approximately 16 cm vertically by continuously moving the sample cell with a fixed radiation source and detectors. NIR spectra were collected by putting the samples into a sample cell (size of window: 3.5 cm × 26 cm) without any pretreatments. Reference single beam spectra of a ceramic standard were collected for every sample immediately before the collection of the sample single beam spectra and the corresponding sample reflectance spectra were obtained. All spectra were recorded as log(1/R) with respect to the ceramic reference standard. Each sample spectrum was obtained by averaging 32 scans from different areas of the sample. Triplicate spectra were collected for each sample.

2.3. Raman spectral collection

A schematic diagram of the WAI scheme is shown in a previous publication [12]. The WAI scheme used in this study was developed by Kaiser Optical Systems (Ann Arbor, MI, USA). An excitation laser (785 nm, diode laser) was magnified to form a circular illumination area with a diameter of 6 mm (area: 28.3 mm^2) in order to cover a large sample area. The scattered radiation was then collected by an array of 50 optical fibers and delivered to the CCD detector (Kaiser RamanRxn1 unit, Kaiser Optical Systems). The pellets were transferred to a glass vial (diameter: 2.54 cm) and located at the focal point of the laser for the measurement.

Triplicate spectra were also collected for each sample. At each spectral collection, a vial containing the sample was shaken vigorously to redistribute the pellets, thereby ensuring that the sample orientation at each triplicate measurement was random. Multiplicative scatter correction (MSC) and PLS were performed using Matlab Version 7.0 (MathWorks Inc., MA, USA).

3. Results and discussion

3.1. Examination of spectral features

Fig. 2(a) shows 25 raw diffuse reflectance NIR spectra (1100–2100 nm) from 25 pellet samples. (One spectrum from each sample is displayed.) The baselines of spectra are vertically varying due to the difference in scattering over the NIR range and differences in sample packing and shape. A certain degree of baseline variation is inescapable in diffuse reflectance measurements of large granular samples, and so we have preprocessed the raw spectra using a multiplicative scatter correction (MSC). MSC is one of the most frequently adopted methods to minimize the unwanted scattering-causing spectral variation present in diffuse reflectance spectra. The resulting MSC-treated spectra are shown Fig. 2(b). The degree of baseline variation is largely decreased after the application of MSC.

Fig. 3(a) shows 25 raw Raman spectra (1504–1054 cm^{-1}) of the same 25 pellet samples whose NIR spectra are shown in Fig. 2. Three of the 25 samples show apparently different backgrounds due to weak fluorescence. One sample from the 1-butene grade and two samples from the 1-octene grade showed the underlying

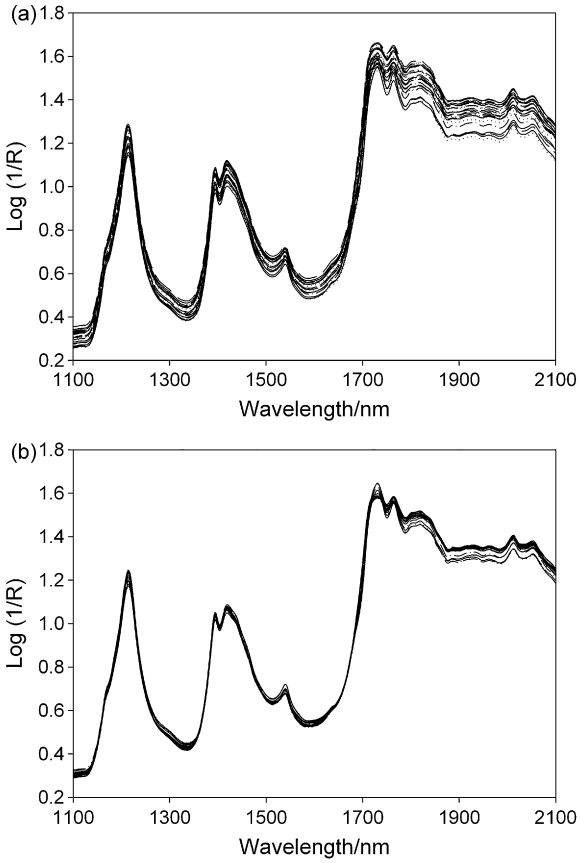

Fig. 2. Twenty-five raw diffuse reflectance NIR spectra from 25 pellet samples (a) and the resulting MSC-treated spectra (b).

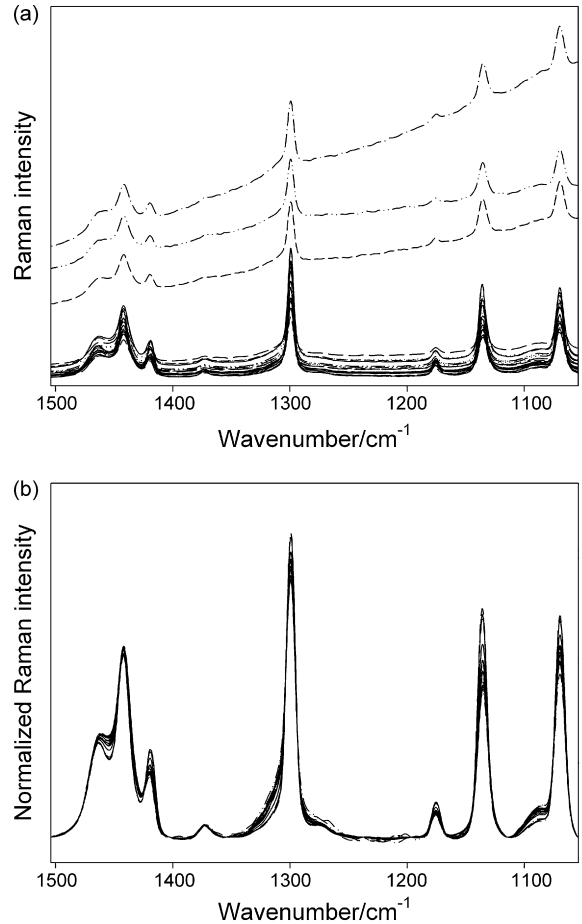

Fig. 3. Twenty-five raw Raman spectra from 25 pellet samples (a) and the resulting normalized spectra (b).

fluorescence backgrounds, and the colors of these three samples were slightly different from the others. The baselines of the other 22 samples varied slightly, but the pattern of baseline variations was fairly consistent. The baselines of the raw Raman spectra were corrected at five different wavenumbers (1504, 1402, 1196, 1114 and 1054 cm^{-1}) and the peak area under the 1504–1054 cm^{-1} range was calculated. Then, each baseline corrected spectrum was divided by the corresponding peak area. Through normalization, the possible Raman peak intensity variations resulting from a change in sample position and/or packing with respect to the focal point were reduced, and the resulting normalized spectra are shown in Fig. 3(b). For further spectral comparison and PLS analysis, the MSC-treated NIR spectra and normalized Raman spectra were used.

Fig. 4(a) shows MSC-treated NIR spectra of two different pellet samples, corresponding to the lowest and highest density in the dataset. The thinner and thicker lined spectra correspond to samples with densities of 0.918 and 0.961 g cm^{-3}, respectively. The bands in the 1100–1320 nm range correspond to the second overtone of C–H stretching. The bands in the 1320–1640 nm range are mostly a combination of the first overtone of C–H stretching and C–H deformation. The first overtone bands of C–H stretching are mostly located in the 1650–2100 nm range. The overall NIR spectral features between the two samples are slightly different throughout the whole spectral range, with the most selective spectral differences observed in the 1700–1900 nm. Since absorptivity generally increases as NIR wavelength increases, a more discernible spectral difference is observed in the first overtone region (1700–1900 nm).

Fig. 4(b) shows normalized Raman spectra of the same pellet samples from Fig. 4(a). As shown and generally known, Raman spectral features are more distinct (narrower bandwidth) and selective compared to NIR. There are eight major Raman bands of polyethylene in the 1504–1054 cm^{-1} range. The 1464, 1443, 1420 and 1372 cm^{-1} bands correspond to CH_2 rocking, CH_2 bending, CH_2 bending and CH_2 wagging, respectively. The 1298 and 1177 cm^{-1} bands are from CH_2 twisting and CH_2 rocking, and both the 1138 and 1070 cm^{-1} bands correspond to C–C (backbone) stretching.

There are fairly significant spectral differences between the two samples in terms of shape as well as intensity of the Raman bands. The higher density sample (thicker line) shows a higher Raman intensity, which is reasonable since, for a sample of higher density, a relatively larger number of molecules should be present in a given volume of laser illumination due to a denser molecular arrangement. The most distinct changes in band shape are observed in the 1500–1400 cm^{-1} range where the information related to CH_2 bending is mostly apparent. The bottom left of the 1298 cm^{-1} band broadens with the lower density sample. A sample of lower density results from a less ordered structural arrangement, while a highly ordered structural arrangement leads to a sample with high density. A low density sample contains more variation in molecular arrangement, which leads to band broadening. Overall, Raman spectral features reflect the density variations more distinctly in correlation with structure.

The two Raman spectra shown in Fig. 4(b) were selected to show the largest spectral variation and easily explain spectral differences caused by density change. When polymers with similar densities are analyzed, the reproducibility of Raman measurements becomes more critical. If the magnitude of the spectral variation from an unsuccessful sample representation or unsatisfactory spec-

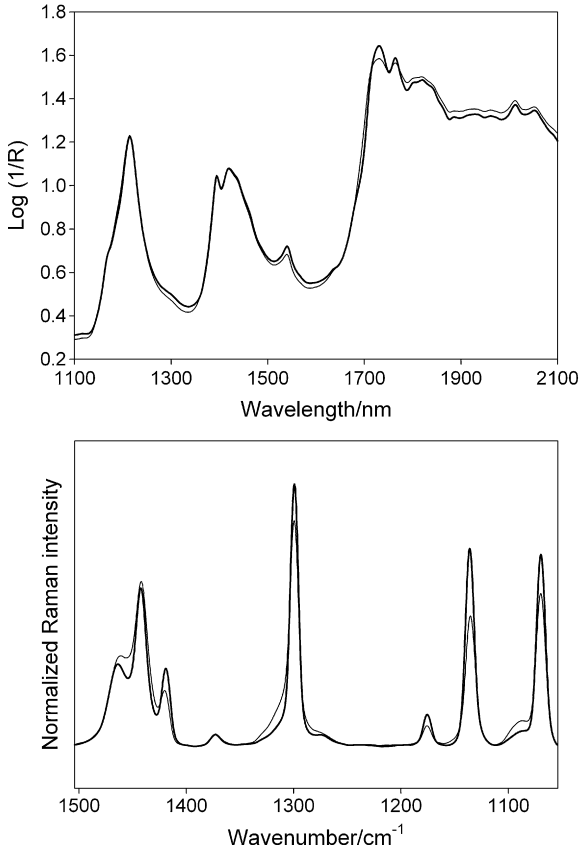

Fig. 4. MSC-treated NIR spectra (top) and normalized Raman spectra (bottom) of two polymer samples with densities of 0.918 (thinner line) and 0.961 g cm^{-3} (thicker line).

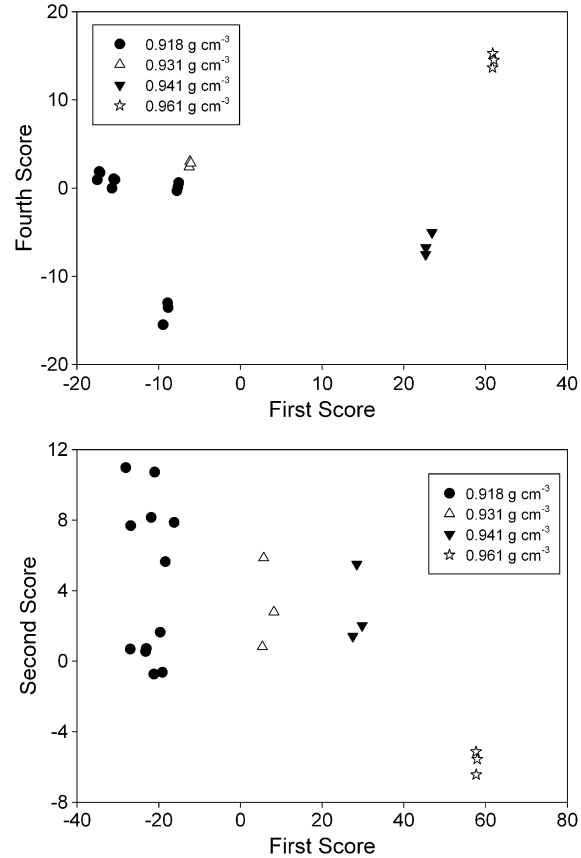

Fig. 5. Score scatter plots of seven arbitrarily selected samples generated from the use of NIR (top) and Raman spectral data (bottom). The first and fourth scores were used for NIR, and the first and second scores were used for Raman to generate two-dimensional score scatter plots.

tral reproducibility is larger than that of the spectral variation from the actual structural differences, the utility of the Raman measurement for quantitative analysis will be greatly diminished, although highly selective information can be obtained, as shown in Fig. 4(b).

3.2. Preliminary investigation of spectral variations using principal component analysis

To preliminarily examine the correlation between spectral variation and the corresponding density change, we have examined the variation of scores generated from principal component analysis (PCA) using seven arbitrarily selected samples. Four of the seven samples had the same density (0.918 g cm^{-3}) with different grades. Using both the NIR and Raman spectra, PCA was performed with the whole spectral range for both cases (1100–2100 nm for NIR, 1504–1054 cm^{-1} for Raman).

Fig. 5 shows the resulting score scatter plots for the selected samples with the use of NIR (top) and Raman spectral data (bottom). The first and fourth scores were used for NIR, and the first and second scores were used for Raman to generate two dimensional score scatter plots. The densities of the seven samples are designated in the figure. The variation of the first score obtained from the Raman spectra is clearly accordant with density variation, and the variation is fairly quantitative (by simple visual examination). The variation of first score obtained from NIR spectra also follows the density change, but it was not as quantitative as the score variation obtained from the Raman spectra. For example, the NIR scores of the sample with a density of 0.931 g cm^{-3} (open triangles) were quite close those of the samples with a density of 0.918 g cm^{-3} (filled circles), and these were much further from the scores of the sample with a density of 0.941 g cm^{-3} (filled triangles). It is convincing that Raman spectral features provide more specific and selective information related to density variation originating from a change of molecular structure and arrangement.

3.3. Using partial least squares to determine the density of polyethylene pellets

To evaluate the accuracy of a given analytical method when multivariate calibration methods such as PLS are used, calibration and validation sets are usually prepared first, and then the resulting errors are evaluated. This approach would be reasonable for judging the accuracy of an analytical method, but the variations in the resulting error depends on how the calibration and validation sets are prepared or divided from the entire dataset. In this study, we have obtained the resulting errors by fully cross-validating all samples in the whole dataset rather than by distinct division into calibration and validation sets. Initially, we fixed the number of samples for the calibration and validation sets at 20 and 5, respectively (total 25 samples). A tremendous number of combinations are possible to select which 5 samples would comprise the validation set ($_{25}C_5$ = 53,130 combinations). At each combination, the standard error of calibration (SEC) and standard error of prediction (SEP) were calculated. Finally, the mean standard error of calibration (MSEC) and the mean standard error of prediction (MSEP) were calculated by averaging the SEC and SEP values obtained from all combinations. The resulting MSEC and MSEP were then used to compare the analytical performance of NIR and Raman

Table 1
Fully cross-validated PLS results for the density measurement using the whole spectral range of NIR and Raman data (1100–2100 nm for NIR, 1504–1054 cm^{-1} for Raman). The numbers in parentheses correspond to the number of factors used.

NIR		Raman	
MSEC ($g\,cm^{-3}$)	MSEP ($g\,cm^{-3}$)	MSEC ($g\,cm^{-3}$)	MSEP ($g\,cm^{-3}$)
0.0016 (5)	0.0028	0.0006 (4)	0.0008

spectroscopy for the determination of the density of polyethylene pellets.

Table 1 presents the fully cross-validated PLS calibration results using the whole spectral range of NIR and Raman data (1100–2100 nm for NIR, 1504–1054 cm^{-1} for Raman). The numbers in parentheses correspond to the number of factors used. The resulting MSEC and MSEP using Raman spectroscopy were substantially superior to those using NIR spectroscopy. As shown in Figs. 4 and 5, Raman spectral features provide more selective chemical information to reflect the structural variations of the polymer; therefore, the resulting MSEP of the Raman measurements is better than that of the NIR measurements. Along with the high selectivity of Raman bands, the improved reproducibility using the WAI scheme for spectral collection also played an important role leading to a more accurate density measurement. The improved reproducibility using the WAI scheme and the higher spectral selectivity of Raman spectroscopy harmonized to improve the accuracy of the density determination. The results in this study are opposite to those from the previous publication, where the accuracy of NIR measurements was better than that of Raman measurements in determining the ethylene content of polypropylene [7]. The less reproducible Raman measurements from the insufficient sample representation caused by probing only localized chemical information and change of sample presentation (such as packing and morphology) was the major cause of the inferior accuracy.

The spectral variation related to the density change should be present through the whole spectral range since density change is related to the variation of all components in a sample rather than a specific component or group of components. However, the use of a narrower spectral range where the related information could be more localized and less immune to instrumental variations may possibly lead to improved calibration results. To find an optimal range for both datasets, we used a moving window PLS (MWPLS) [18–20]. The main strategy of the MWPLS was to perform PLS at many different spectral ranges by changing the size and location of the spectral window, thus determining the spectral range that yields the minimal error. To optimize the NIR spectral range, the window size varied from 50 to 1000 nm in 25 nm increments. For every given window size, the window shifted from a low to a high wavelength in 25 nm steps until the whole spectral range (1100–2100 nm) was covered. This method enabled a total of 780 different NIR ranges to be evaluated. Similarly, to optimize the Raman spectral range, the window size varied from 50 to 450 cm^{-1} in 25 cm^{-1} increments, yielding a total of 153 different Raman ranges. In each spectral range (window), the MSEC and MSEP were obtained by full cross-validation as described above and finally the optimal ranges for both the NIR and Raman datasets were identified.

Table 2 summarizes the observed optimal ranges for both methods when anywhere from one to five factors were used. As shown, the best NIR accuracy was achieved using the 1975–2100 nm range with five factors. By using this optimal NIR spectral range, the resulting MSEP was significantly improved compared to the use of the whole spectral range shown in Table 1. The use of the 1454–1054 cm^{-1} range with four factors led to the best accuracy for Raman measurements, and the MSEP using this optimized range was the same as using the whole spectral range. Although NIR calibration accuracy was improved dramatically by spectral range optimization, Raman accuracy is still better for the determination of the density of polyethylene pellets.

Table 2
The found optimal spectral ranges and the corresponding MSEC and MSEP values when different numbers of factors are used.

	Number of factor	MSEC ($g\,cm^{-3}$)	MSEP ($g\,cm^{-3}$)	Optimal range
NIR	1	0.0027	0.0029	1225–1750 nm
	2	0.0021	0.0025	1425–1775 nm
	3	0.0014	0.0020	1525–1725 nm
	4	0.0010	0.0014	1950–2100 nm
	5	0.0008	0.0011	1975–2100 nm
Raman	1	0.0011	0.0011	1479–1129 cm^{-1}
	2	0.0009	0.0010	1479–1104 cm^{-1}
	3	0.0007	0.0009	1454–1104 cm^{-1}
	4	0.0006	0.0008	1454–1054 cm^{-1}
	5	0.0006	0.0008	1454–1054 cm^{-1}

In the case of NIR range optimization, the optimized ranges varied considerably when the number of factors used for optimization changed, a fact which did not hold for Raman spectroscopy. The Raman spectral features were selective across the whole spectral range, as shown Fig. 4; therefore, the optimized ranges did not vary greatly depending on the number of factors used. For the case of NIR, the most selective bands in the 1700–1800 nm range were partially included up to three factors, with a resulting MSEP of 0.0020 $g\,cm^{-3}$. Using 5 factors, the MSEP improved substantially to 0.0011 $g\,cm^{-3}$, but the optimized range was significantly narrowed (to 1975–2100 nm). The 1975–2100 nm range should provide the spectral variation related to density change, but it might not be the most selective range based on the examination of the two NIR spectra shown in Fig. 4. Although accurate NIR density measurements are reachable by optimizing a spectral range for PLS, the sensitive variation of accuracy depending on the number of factors used raises possible question on the robustness of NIR measurements. When a minor NIR spectral variation is introduced (due to a change in instrumental conditions), the found optimal range would be no longer optimal, which may increase the prediction error.

Fig. 6 shows the prediction correlation plots obtained from the NIR (top) and Raman data (bottom). The prediction results using the 1975–2100 nm range with 5 factors for NIR and the 1454–1054 cm^{-1} range with 4 factors for Raman are plotted in the figure. Clearly, the correlation between reference density and predicted density is better when Raman spectroscopy is employed. The resulting R^2 values using NIR and Raman spectroscopy were 0.9948 and 0.9969, respectively.

Additionally, to compare the repeatability of NIR and Raman measurements, we calculated the standard deviation of three predicted values from each sample and the average standard deviation was obtained from the 25 standard deviations calculated from the 25 samples. The average standard deviations for NIR and Raman predictions were 0.00033 and 0.00030 $g\,cm^{-3}$, respectively. The resulting repeatabilities of both methods were nearly identical. Although Raman spectra were collected without sample rotation or reciprocation during spectral collection and a vial containing a sample was shaken vigorously to redistribute the pellets, the same level of repeatability was achieved as when the NIR spectra were collected by covering large sample areas. Based on the results of these repeatability comparisons, Raman spectra collected with the use of the WAI scheme are repeatable and reproducible in their representation of the overall chemical identity of a sample.

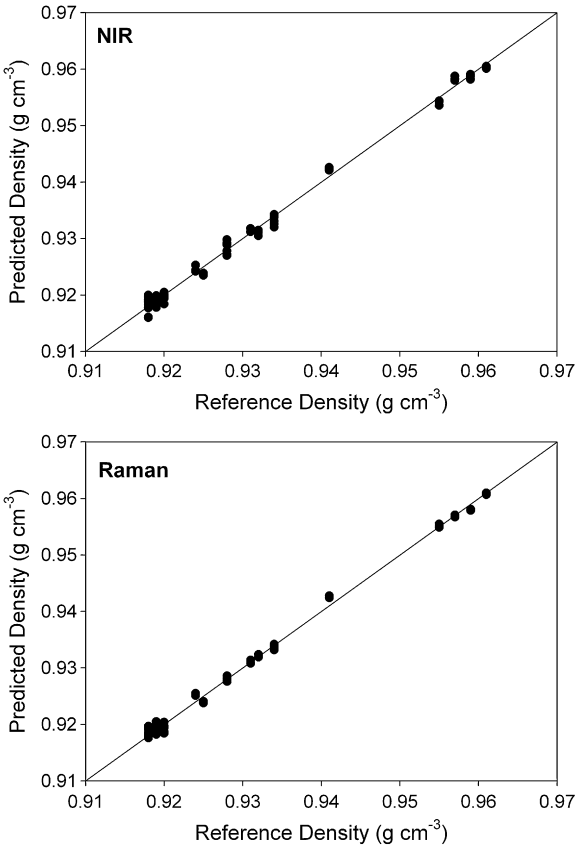

Fig. 6. Prediction correlation plots obtained from NIR (top) and Raman data (bottom). The prediction results were obtained using the 1975–2100 nm range with five factors for NIR and the 1454–1054 cm^{-1} range with four factors for Raman.

An excellent Raman prediction accuracy of 0.0008 $g\,cm^{-3}$ was obtained using the WAI scheme for density determination as compared to previous reports of 0.0022 $g\,cm^{-3}$ for polyethylene [9], 0.0018 $g\,cm^{-3}$ for linear low-density polyethylene [10] and 0.0024 $g\,cm^{-3}$ for poly(ethylene terephthalate) [11]. Although it might not be pertinent to directly compare the accuracy obtained in this study to those from other studies since the characteristics of sample, the used Raman spectroscopic system and the strategy of chemometric analysis were all different, we have sufficiently demonstrated that Raman spectral collection using the WAI scheme has strong potential to improve the accuracy of quantitative analysis for diverse polymer samples.

4. Conclusions

The combination of selective Raman features and reproducible spectral collections with correct sample representations using the WAI scheme led to a highly accurate determination of the density of polyethylene pellets. Although Raman spectra were collected without sample rotation or reciprocation, the repeatability was comparable to that of diffuse reflectance NIR spectroscopy. If sample rotation for the WAI Raman measurement was permitted, the resulting accuracy should be further improved (an expectation which we expect to verify in the near future). Even though there are long-lasting arguments on the pros and cons of using NIR or Raman spectroscopy, the results presented in this study promote Raman spectroscopy over NIR for the quantitative analysis of diverse polymer samples.

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